Lingjun Liu

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EDUCATION

North Carolina State University (NCSU) — **August 2024 - May 2028**
Ph.D., Computer Science (GPA: 4.0/4.0) — *Raleigh, NC, USA*

Korea Advanced Institute of Science and Technology (KAIST) — **September 2019 - August 2021**
M.S., Computer Science (GPA: 3.53/4.3) — *Daejeon, South Korea*

Korea Advanced Institute of Science and Technology (KAIST) — **August 2018 - June 2019**
Exchange Program, Computer Science — *Daejeon, South Korea*

National Tsing Hua University — **September 2014 - June 2019**
B.S., Computer Science (GPA: 3.75/4.3) — *Hsinchu, Taiwan*

SKILLS

Languages: Python, C++, Java, R
Tools & Frameworks: Git, Docker, REST API, Java Spring, MongoDB, SLURM, Linux/Unix, Advanced Installer, LangChain
CI/CD: Jenkins, GitHub Actions
Testing & Systems: Mutation testing, differential testing, fuzzing, static analysis, SMT solvers (Yices, Z3), CI/regression testing

EXPERIENCE

North Carolina State University
Graduate Research Assistant — *Raleigh, NC, USA* · **August 2024 - Present**

- Built a data pipeline that mines 1,760 historical GCC/LLVM bug reports via GitHub API and HTML parsing into structured test-case pairs, powering IssueMut, a bug-detection framework for C compilers (published, MSR 2026)
- Designed a multi-agent LLM pipeline (LangChain, Gemini) - generation, validation, and iterative-revision agents - to synthesize 587 fuzzing mutators from the mined test cases; found 65 bugs (60 confirmed/fixed) by GCC/LLVM maintainers
- Integrated mined mutators into two leading mutational fuzzers (MetaMut and Kitten); the augmented variants discovered up to 1.9× more unique crashes than the original baselines over 24-hour fuzzing campaigns
- Mentored undergraduate researchers on mutator scripting and compiler bug reporting, accelerating expansion of the mutator library for IssueMut and shortening the time from bug discovery to reporting
- Co-led a differential-testing pipeline for deep-learning compilers, parallelized via SLURM job scheduling to run multiple test campaigns concurrently; found 76 bugs across PyTorch, JAX, TensorFlow, and XLA, 45 confirmed or fixed

Suresoft Technologies Inc.
Associate Research Engineer — *Seongnam, South Korea* · **November 2022 - July 2023**

- Shipped production C++ rule checkers (AUTOSAR C++14) to a 200+ rule suite as a core member of a 3-person team (later 5-7) in a commercial static-analysis product used by paying customers; grew from new hire to core contributor within 2 months, boosting rule-checker throughput 3x (1–2/week → 5/week)
- Built a backend license-validation service with Java Spring that enforced time-limited licensing for commercial customers, ensuring compliance and preventing unauthorized use
- Designed and built a rule-description generator using Java Spring and MongoDB, exposing a REST API for frontend data retrieval and creating the database schema, which streamlined access to rule metadata for internal tools
- Onboarded and mentored two new engineers through pair-programming and code-preview sessions, accelerating their readiness for formal reviews and improving team productivity
- Managed software packaging with Advanced Installer and collaborated with QA to debug and resolve packaging-related test failures
- Documented edge cases and rule checking logic in Jira for code review; supported regression testing across web-based product releases

KAIST
Full-time Researcher — *Daejeon, South Korea* · **November 2023 - April 2024**

- Migrated the reliability computation engine from R/WinBUGS/Shiny to PyMC to align with the Python-based backend, enabling seamless integration with the rest of the system; Built out the pipeline for updating reliability estimates dynamically from incoming test outcome data
- Containerized a full-stack application (Next.js, FastAPI, MySQL) with Docker and Docker Compose, and built a one-command deployment script to automate multi-container builds

Full-time Researcher — *Daejeon, South Korea* · **October 2021 - August 2022**

- Evaluated and prototyped statistical modeling approaches (Bayesian Belief Network) for a reliability computation engine in nuclear safety-critical software, implementing the first version in R/WinBUGS with a Shiny dashboard for visualizing reliability estimates
- Designed the calculation logic that determines test counts required to achieve target reliability thresholds based on pass/fail outcomes

Graduate Research Assistant — *Daejeon, South Korea* · **August 2018 - August 2021**

- Developed MuFBDTester, an automated test generator for safety-critical nuclear reactor protection system using mutation testing and SMT constraint solving (Yices); achieved 100% detection of real industrial faults, cut manual test-suite creation from days to minutes, and ran 20x faster than leading structural-coverage tools (published, STVR 2022)
- Modeled multi-agent attack scenarios and generated test cases via model checking for air-traffic-control-system security testing

AWARDS & FELLOWSHIPS

- University Graduate Fellowship at NCSU - Top graduate student fellowship - Aug 2024
- Best Short Paper Award at the Korea Conference on Software Engineering - Feb 2024
- Best Artifact Award at the Symposium on Software Engineering for Adaptive and Self-Managing Systems - May 2021